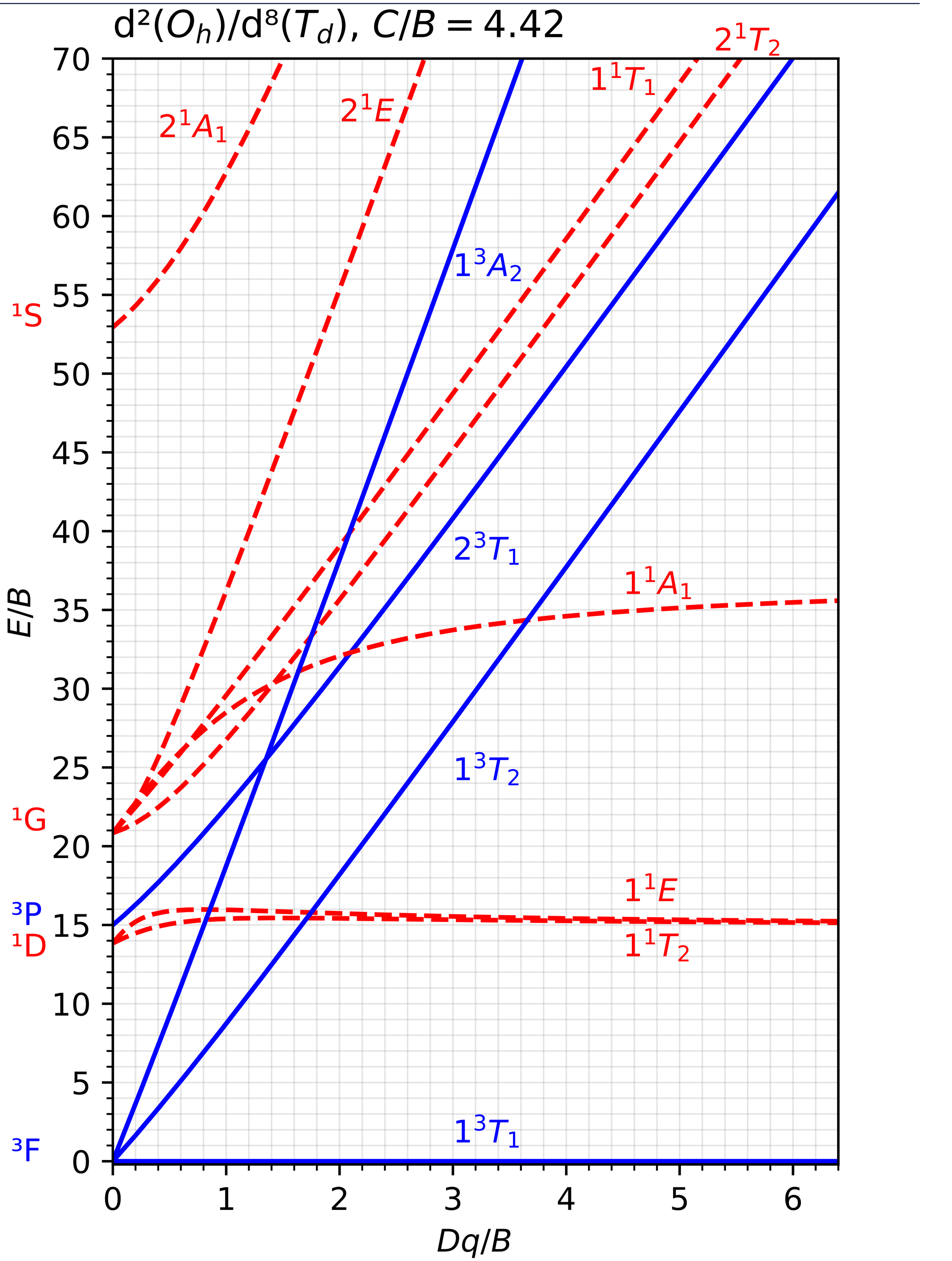
Tanabe–Sugano Diagram



Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\,\Sigma_{i<j}\, r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$

$^1A_1(^1\mathrm{G}, {}^1\mathrm{S})$:

t_2^2	e^2
$10B+5C$	$\sqrt{6}(2B+C)$
	$8B+4C$

$^1E(^1\mathrm{D}, {}^1\mathrm{G})$:

t_2^2	e^2
$B+2C$	$-2\sqrt{3}B$
	$2C$

$^1T_2(^1\mathrm{D}, {}^1\mathrm{G})$:

t_2^2	$t_2{}^1e^1$
$B+2C$	$2\sqrt{3}C$
	$2C$

$^3T_1(^3\mathrm{F}, {}^3\mathrm{P})$:

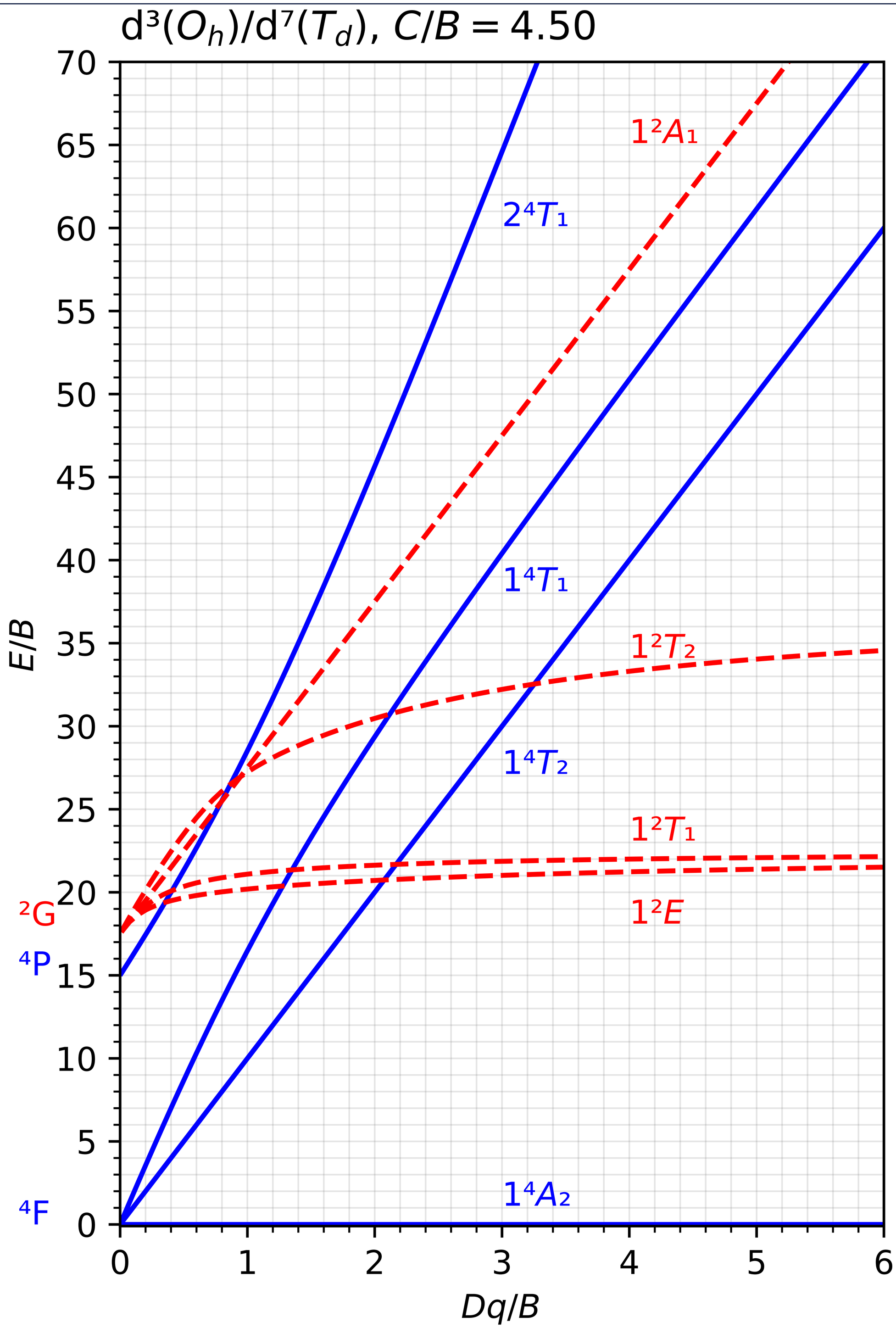
t_2^2	$t_2^1e^1$
$-5B$	$6B$
	$4B$

$^1T_1(^1\mathrm{G})$: $t_2^1e^1,\ 4B+2C$

$^3T_2(^3\mathrm{F})$: $t_2^1e^1,\ -8B$

$^3A_2(^3\mathrm{F})$: $e^2,\ -8B$

Tanabe–Sugano Diagram



Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\,\Sigma_{i<j}\, r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$

$^2E(1^2\mathrm{D}, 2^2\mathrm{D}, {}^2\mathrm{G}, {}^2\mathrm{H})$:

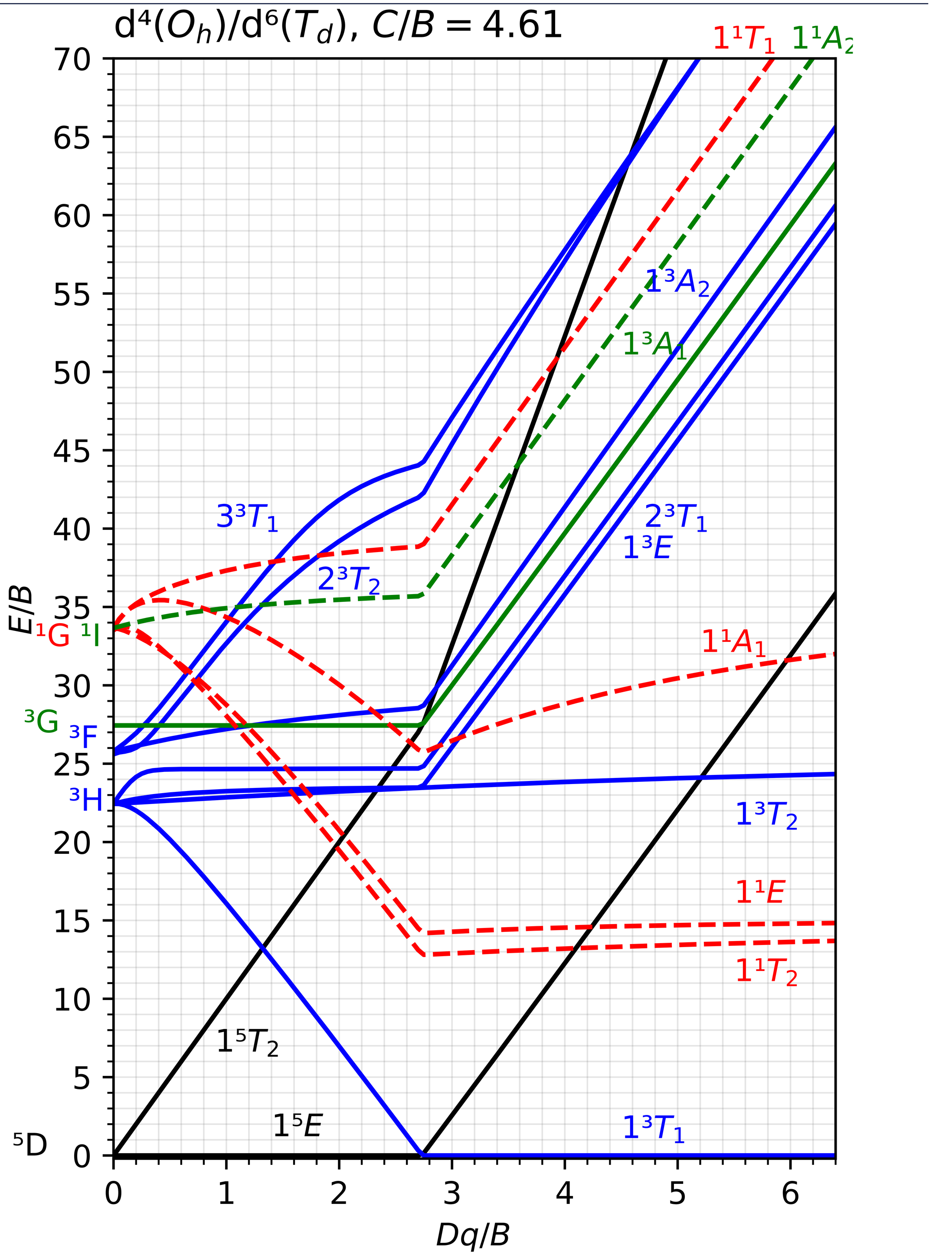
t_2^3	$t_2^2(^1A_1)e^1$	$t_2^2(^1E)e^1$	e^3
$-6B+3C$	$-6\sqrt{2}B$	$-3\sqrt{2}B$	0
	$8B+6C$	$10B$	$\sqrt{3}(2B+C)$
		$-B+3C$	$2\sqrt{3}B$
			$-8B+4C$

$^4T_1(^4\mathrm{P}, {}^4\mathrm{F})$:

$t_2^2(^3T_1)e^1$	$t_2^1e^2(^3A_2)$
$-3B$	$6B$
	$-12B$

$^4A_2(^4\mathrm{F})$: $t_2^3,\ -15B$
$^4T_2(^4\mathrm{F})$: $t_2^2(^3T_1)e^1,\ -15B$
$^2A_1(^2\mathrm{G})$: $t_2^2(^1E)e^1,\ -11B+3C$
$^2A_2(^2\mathrm{F})$: $t_2^2(^1E)e^1,\ 9B+3C$

Tanabe–Sugano Diagram



Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\sum_{i<j} r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$

$^3A_2(^1\mathrm{F}, ^2\mathrm{F})$:

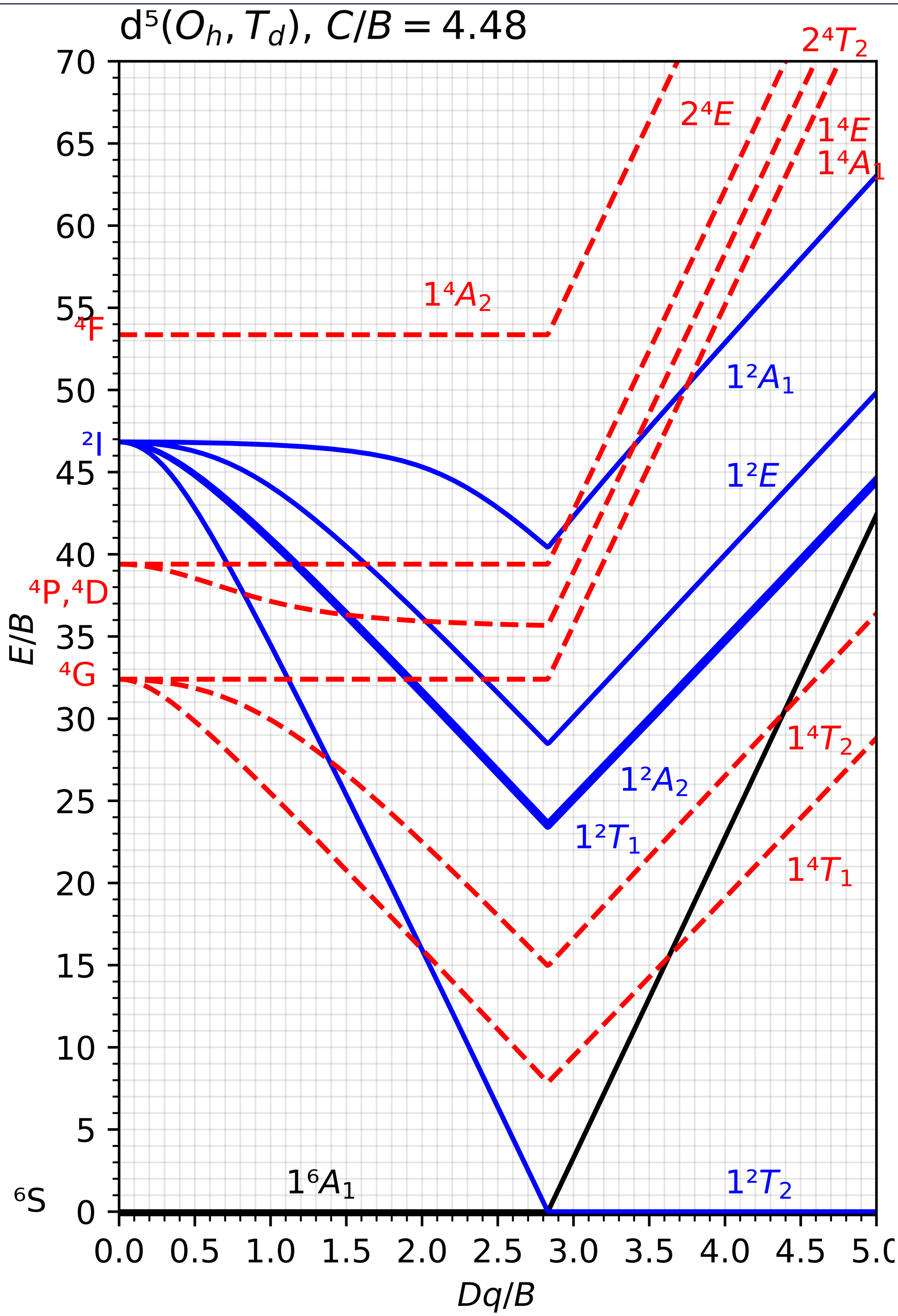
$t_2^3(^2\mathrm{E})e^1$	$t_2^2(^1\mathrm{A}_1)e^2(^3A_2)$
$-8B+4C$	$-12B$
	$-2B+7C$

$^1A_2(^1\mathrm{F}, ^1\mathrm{I})$:

${t_2}^3(^2\mathrm{E})e^1$	${t_2}^2(^1\mathrm{E})e^2(^1\mathrm{E})$
$-12B+6C$	$6B$
	$-3B+6C$

$^5E(^5\mathrm{D}) : t_2^3(^4A_2)e, \ -21B$
$^5T_2(^5\mathrm{D}) : t_2^2(^3T_1)e^2(^3A_2), \ -21B$
$^3A_1(^3\mathrm{G}) : t_2^3(^2E)e, \ -12B+4C$

Tanabe–Sugano Diagram



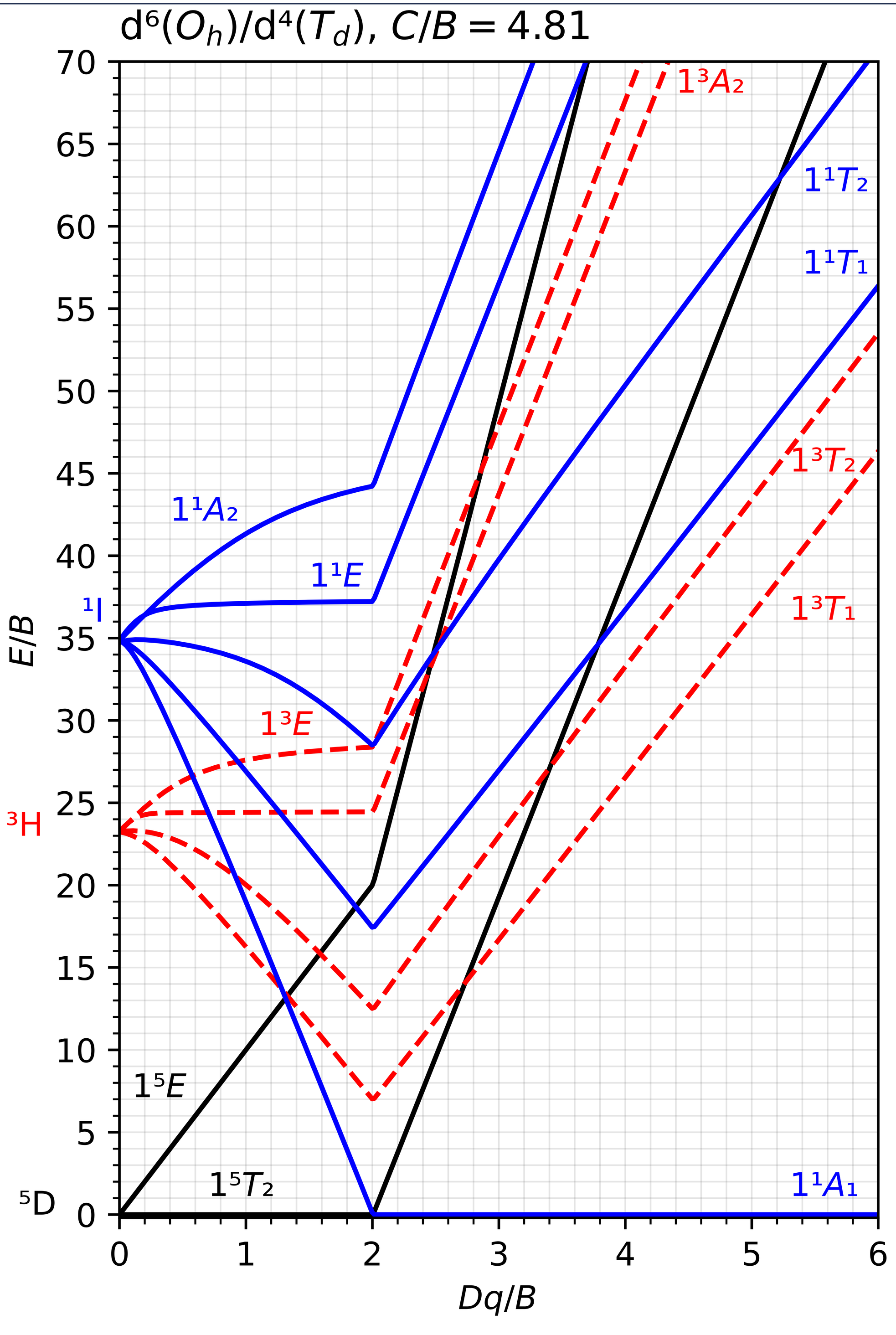
Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\sum_{i<j} r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$

$^4E(^4\mathrm{D}, ^4\mathrm{G})$:

$t_2^3(^2\mathrm{E})e^2(^3A_2)$	$t_2^3(^4\mathrm{A}_2)e^2(^1E)$
$-22B+5C$	$-2\sqrt{3}B$
	$-21B+5C$

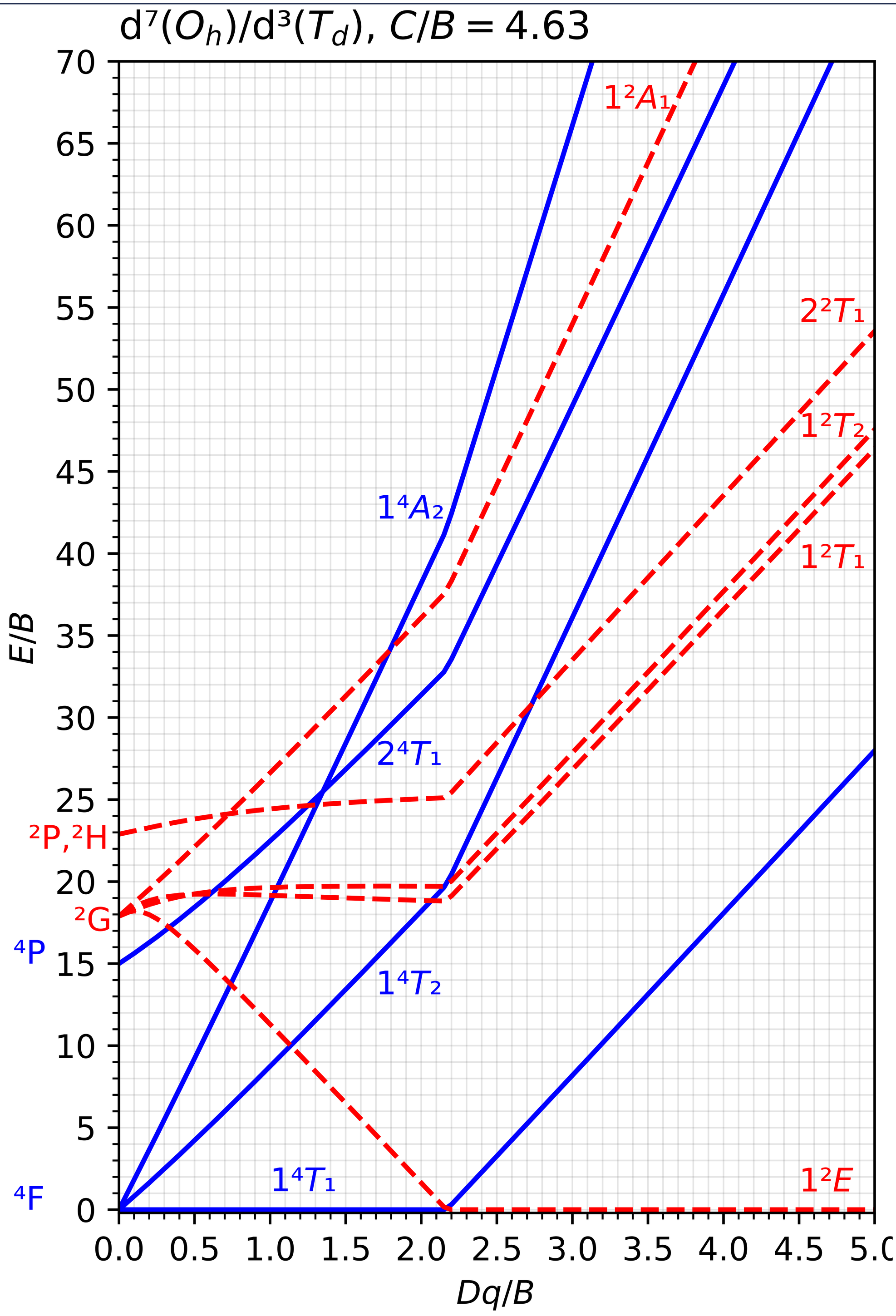
$^6A_1(^6\mathrm{S}) : t_2^3(^4A_2)e^2(^3A_2), \quad -35B$
$^4A_1(^4\mathrm{G}) : t_2^3(^4A_2)e^2(^3A_2), \quad -25B+5C$
$^4A_2(^4\mathrm{F}) : t_2^3(^4A_2)e^2(^1A_1), \quad -13B+7C$

Tanabe–Sugano Diagram



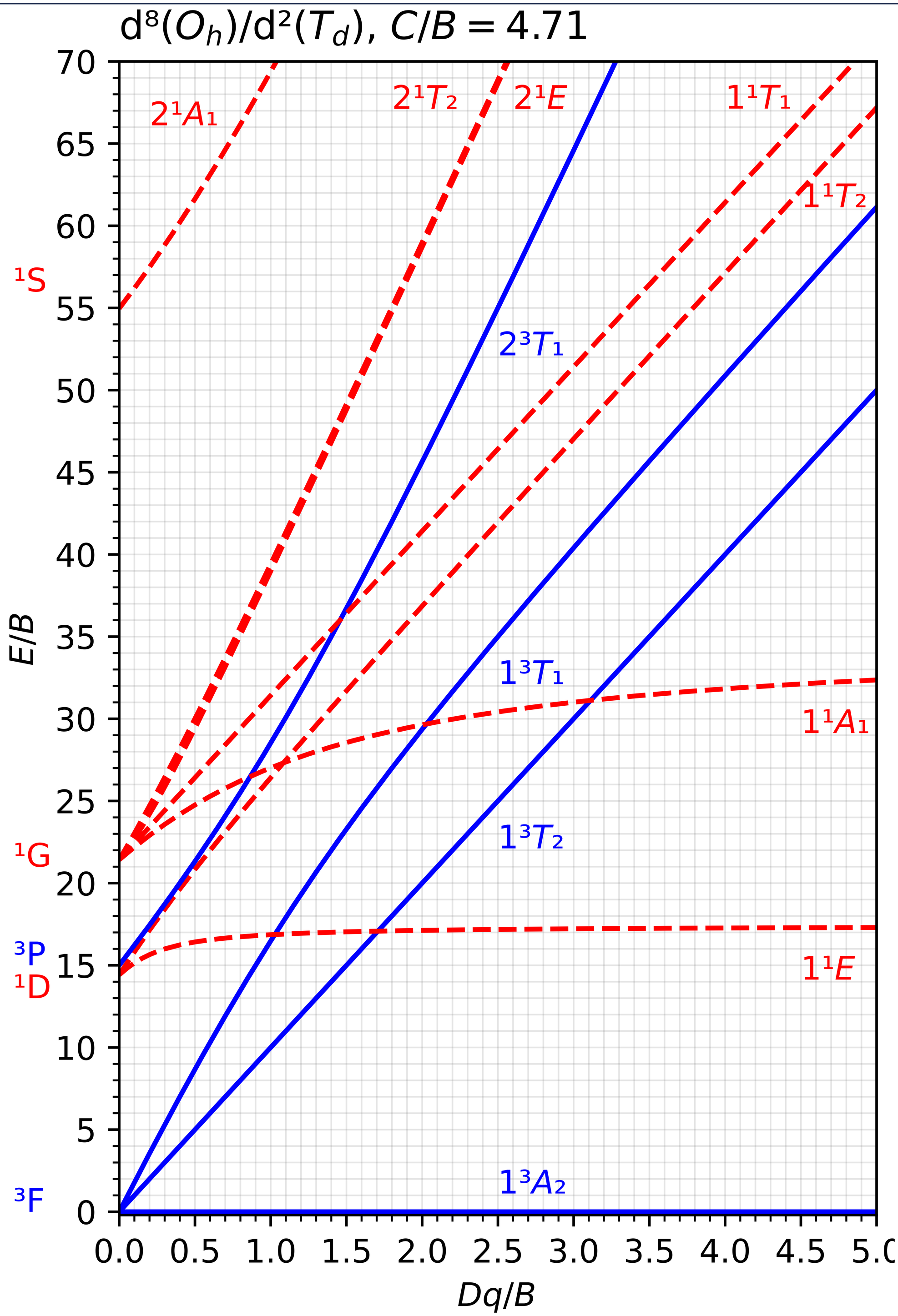
Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\sum_{i<j} r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$

Tanabe–Sugano Diagram



Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\sum_{i<j} r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$

Tanabe–Sugano Diagram



Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\sum_{i<j} r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$